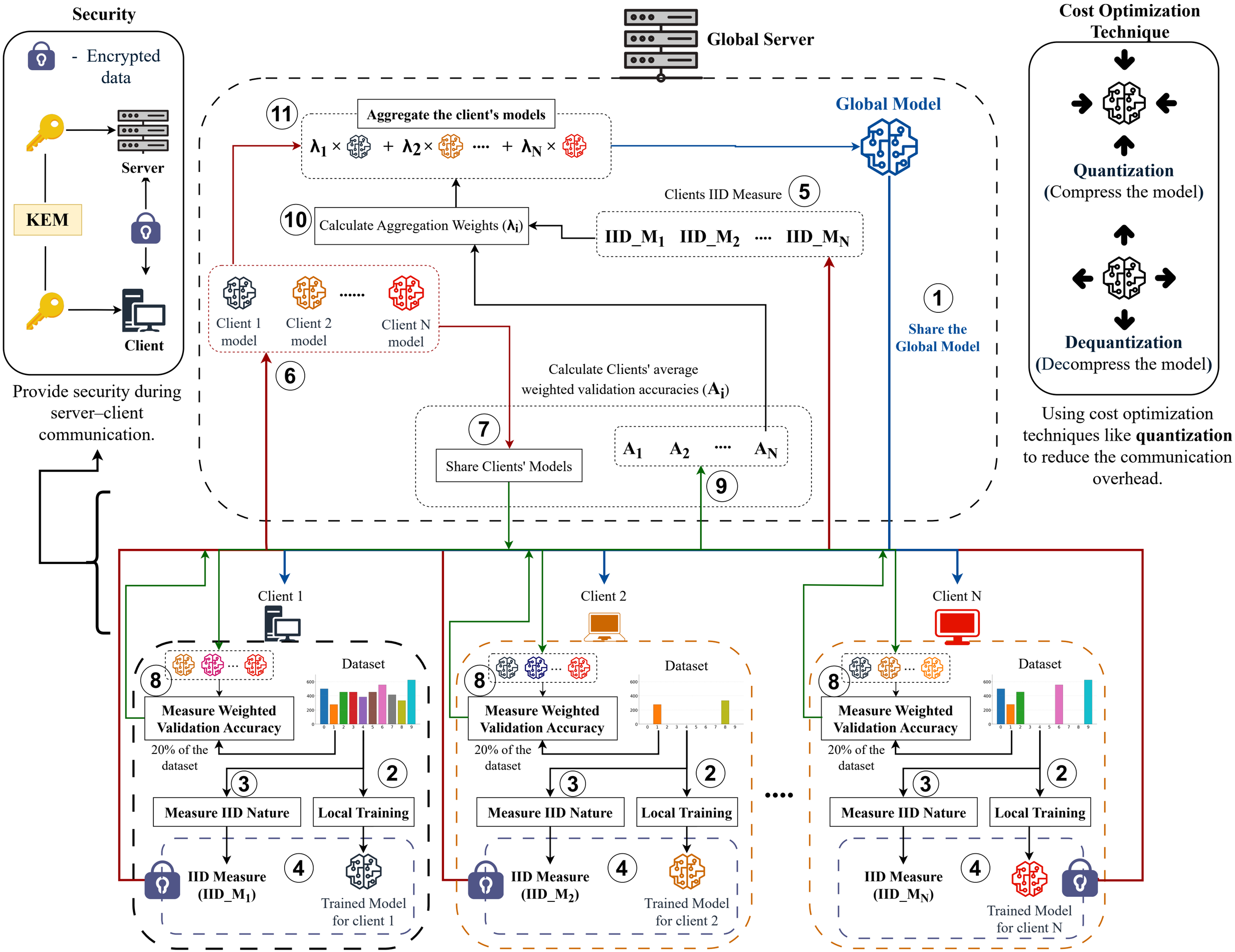
Security
- Encrypted data
Server
KEM
Client
Provide security during server–client communication.
Global Server
Global Model
Aggregate the client's models
λ1 × + λ2 × .... + λN ×
Calculate Aggregation Weights (λi)
Clients IID Measure
IID_M1 IID_M2 .... IID_MN
Client 1 model
Client 2 model
.....
Client N model
Calculate Clients' average weighted validation accuracies (Ai)
Share Clients' Models
A1 A2 .... AN
Share the Global Model
Cost Optimization Technique
Quantization
(Compress the model)
Dequantization
(Decompress the model)
Using cost optimization techniques like quantization to reduce the communication overhead.
Client 1
Client 2
Client N
Dataset
Measure Weighted Validation Accuracy
20% of the dataset
Measure IID Nature
Local Training
IID Measure (IID_M1)
IID Measure (IID_M2)
IID Measure (IID_MN)
Trained Model for client 1
Trained Model for client 2
Trained Model for client N